ICAIN-2026
International Conference on Artificial Intelligence and Networking

26th -28th November 2026
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Indian Institute of Information Technology, Allahabad

************** CALL FOR PAPERS ***************

SPECIAL SESSION ON

Machine Learning and GenAI for Intelligent Systems and Decision Making

SESSION ORGANIZERS:

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EDITORIAL BOARD: (Optional)

[Name, University or Organization, Country, e-mail]

SESSION DESCRIPTION:

The rapid convergence of Machine Learning (ML) and Generative Artificial Intelligence (GenAI) is redefining the boundaries of what intelligent systems can perceive, reason about, and decide. From autonomous agents navigating complex environments to language models augmenting human judgment in high-stakes domains, these technologies are reshaping how decisions are made across science, engineering, business, healthcare, and public policy. This special session provides a dedicated forum to examine both the theoretical foundations and practical frontiers of ML and GenAI as enablers of intelligent, adaptive decision-making systems.

The session welcomes a broad spectrum of contributions spanning deep learning architectures, reinforcement learning, large language and multimodal models, probabilistic reasoning, and human-AI collaboration applied to real-world decision-support and autonomous system challenges. By bringing together researchers from academia and industry across diverse application domains, the session aims to foster cross-disciplinary dialogue on how intelligent systems can be made more capable, efficient, reliable, and trustworthy.

Original research papers, survey articles, and position papers addressing open problems, novel methodologies, benchmark evaluations, and deployment experiences are all welcome. The session particularly encourages work that bridges the gap between cutting-edge ML/GenAI methods and their integration into robust, interpretable, and ethically sound decision-making pipelines in real-world settings.

RECOMMENDED TOPICS:

Topics to be discussed in this special session include (but are not limited to) the following:

- **Deep Learning Architectures for Intelligent Systems —** transformer models, graph neural networks, hybrid neuro-symbolic architectures, and efficient model design for real-time decision pipelines.
- **Generative AI for Knowledge Synthesis and Decision Support —** large language models (LLMs), retrieval-augmented generation (RAG), and GenAI-driven tools for reasoning, summarization, and expert augmentation.
- **Reinforcement and Multi-Agent Learning —** deep reinforcement learning, multi-agent coordination, reward shaping, and autonomous agents for sequential decision-making under uncertainty.
- **Multimodal Learning and Perception —** vision-language models, audio-visual fusion, sensor integration, and cross-modal reasoning for intelligent situational awareness and response.
- **Explainable and Trustworthy AI** — interpretable ML models, post-hoc explanation methods, uncertainty quantification, and human-AI interaction design for accountable decision systems.
- **Transfer Learning, Few-Shot, and Zero-Shot Methods —** domain adaptation, meta-learning, foundation model fine-tuning, and low-resource learning strategies for generalizable intelligent systems.
- **ML for Optimization and Operations Research —** learning-to-optimize, combinatorial decision-making, ML-assisted simulation, and AI for resource allocation and scheduling problems.
- **Federated and Distributed Learning —** privacy-preserving collaborative ML, communication-efficient training, heterogeneous data challenges, and decentralized intelligent systems.
- **GenAI for Synthetic Data and Simulation —** data augmentation, digital twins, scenario generation, and simulation-driven training for robust intelligent system development.
- **Responsible AI and Ethical Decision Making —** fairness, bias mitigation, regulatory compliance, value alignment in GenAI, and societal impact of autonomous decision-making systems.
- **Applications of ML and GenAI in Intelligent Systems** — spanning healthcare, smart cities, autonomous vehicles, finance, education, robotics, supply chains, and scientific discovery.

SUBMISSION PROCEDURE:

Researchers and practitioners are invited to submit papers for this particular theme session on **Machine Learning and GenAI for Intelligent Systems and Decision Making** ***on or before*** **30th August 2026.** All submissions must be original and may not be under review by another publication. INTERESTED AUTHORS SHOULD CONSULT THE CONFERENCE'S GUIDELINES FOR MANUSCRIPT SUBMISSIONS at https://www.icain-conf.com/downloads. All submitted papers will be reviewed on a double-blind, peer-review basis.

NOTE: While submitting a paper in this special session, please specify **Machine Learning and GenAI for Intelligent Systems and Decision Making** at the top (above paper title) of the first page of your paper.

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